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Palaeomagnetic evidence for Tertiary counterclockwise rotation of Adria

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Abstract

With the aim of obtaining Tertiary palaeomagnetic directions for the Adriatic Foreland of the Dinaric nappe system, we carried out a palaeomagnetic study on platform carbonates from stable Istria, from the northwestern and the Central Dalmatia segment of imbricated Adria. Despite the weak to very weak natural remanences of these rocks, we obtained tectonically useful palaeomagnetic directions for 25 sites from 20 localities. All exhibit westerly declinations, both before and after tilt correction. Concerning the age of the magnetizations, we conclude that five subhorizontal and magnetite bearing Eocene localities from stable Istria are likely to carry primary remanence, whereas three tilted and hematite-bearing ones were remagnetized. In the northwestern segment of imbricated Adria the cluster of the mean directions improved after tectonic correction indicating pre-tilting magnetization. In contrast, Maastrichtian–Eocene platform carbonates from Central Dalmatian were remagnetized in connection with the late Eocene–Oligocene deformation or Miocene hydrocarbon migration. Based on the appropriate site/locality means, we calculate mean palaeomagnetic directions for the above three areas and suggest an alternative interpretation of the data of Kissel et al. [J. Geophys. Res. 100 (1995) 14999] for the flysch of Central Dalmatia. The four area mean direction define a regional palaeomagnetic direction of Dec = 336°, Inc = + 52°, $k = 107$, $\alpha_{95} = 9°$. From these data we conclude that stable Istria, in close coordination with imbricated Adria, must have rotated by 30° counterclockwise in the Tertiary, relative to Africa and stable Europe. We suggest that the latest Miocene–early Pliocene counterclockwise rotations observed in northwestern Croatia and northeastern Slovenia were driven by that of the Adriatic Foreland, i.e. the rotation of the latter took place between 6 and 4 Ma.
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Keywords: Tertiary; Palaeomagnetism; Istria; Adria; Rotation

1. Introduction

The present shape and internal structure of the Alpine belt is controlled by the geometry of the converging European and African plates and the movements of the Tethyan blocks, which are lithospheric fragments of high internal coherence. In the central part of the European sector of the belt, the largest and most important area of relative stability is the Adriatic Foreland. This has been considered by palaeomagnetists as an African promotory (e.g. Zijderveld et al., 1970; Soffel, 1972; Chanell and Hor-

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váth, 1976; Chanell et al., 1979; Channell, 1996; Muttoni et al., 2001) or as a microplate (e.g. Márton and Veljović, 1983; VanderBerg, 1983; Márton et al., 1995). Different interpretations were based on palaeomagnetic data from Mesozoic (mainly Cretaceous) rocks and were possible because: (1) scarce and loosely dated palaeomagnetic results from Africa did not provide sufficiently well constrained reference directions during the fast Cretaceous apparent polar wander (Márton, 1993); (2) some data were not free from present field overprints (compare Channell, 1977; VandenBerg, 1983); (3) palaeomagnetic data from deformed units, like the Southern Alps (e.g. Schonborn, 1990; Roeder, 1992; Holliger and Kissling, 1992; Channell and Doglioni, 1994; Bertotti et al., 1998; Devoti et al., 2002) were regarded as representative of Adria (Channell, 1996; Muttoni et al., 2001).

Rotation of Africa slows down in the Tertiary (Besse and Courtillot, 1991). Thus, the reference framework offers a better chance for correct interpretation. That is why the present study deals with uppermost Maastrichtian–Eocene sediments which are biostratigraphically well-dated and therefore, the lower age limit of the acquisition of the remanent magnetization is well constrained. We sampled Eocene rocks from stable Istria (unquestionably part of the Adriatic Foreland; Fig. 1, area 1) and from its margin (Fig. 1, areas 2 and 3). In Central Dalmatia, which includes the mainland, Pelješac peninsula and the islands in the Split–Makarska segment (Fig. 1, area 4), we collected samples from mostly uppermost Maastrichtian strata. The reason is that Eocene limestones were usually weathered or full of larger foraminifers and the flysch overlying the platform (area 5, Fig. 1) was earlier studied by

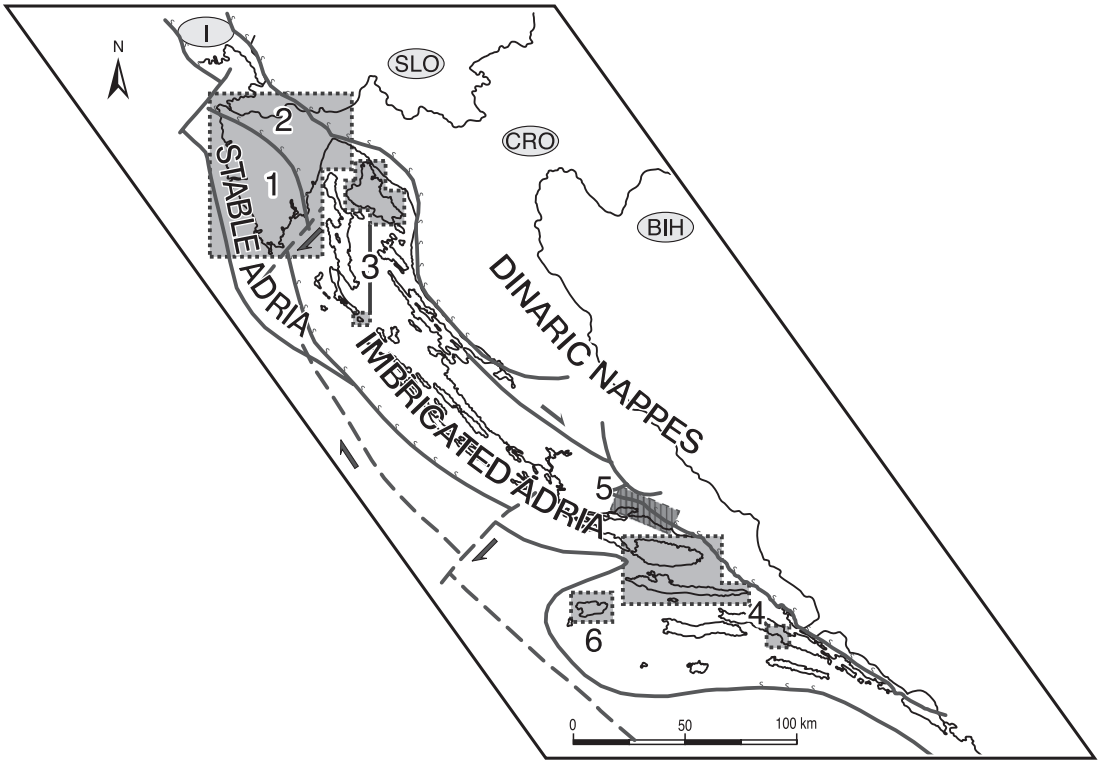


Fig. 1. Sketch map of the Eastern Adriatic coast, modified after Tari (2002). The areas of the present study are: 1 = stable Istria; 2 = imbricated Adria north of Istria; 3 = imbricated Adria of Kvarner islands (Krk and Mali Lošinj); 4 = imbricated Adria of Central Dalmatia; 5 = study area of Kissel et al. (1995); 6 = study area of Márton and Miličević (1994).

Kissel et al. (1995). Nevertheless, we visited the localities sampled by Kissel et al. (1995), in order to compare the degree and style of tectonization of the flysch and the underlying platform. This was important since Kissel et al. (1995) suggested that the flysch did not rotate with respect to stable Europe, while palaeomagnetic data from Cretaceous rocks from the nearby Vis island (Fig. 1, area 6, Márton and Miličević, 1994) as well as the results of the present study from Central Dalmatia do indicate rotation.

2. Geological setting

Numerous studies have dealt with the present-day structural architecture, tectonic evolution and palaeogeography of the Adriatic region (e.g. Kossmat,

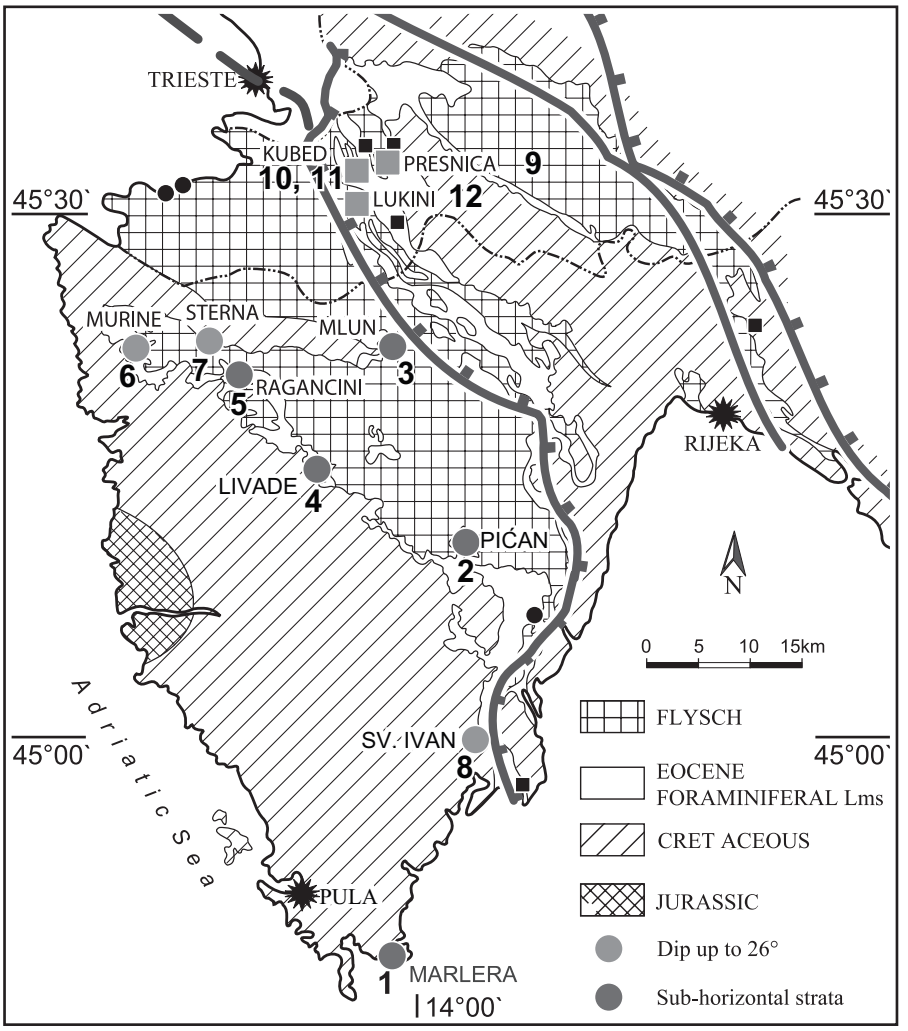


Fig. 2. Geological sketch map of the stable Istria and imbricated Adria north of stable Istria (areas 1–2) after Basic Geological map of former Yugoslavia 1:500 000) with studied localities (dots, squares and numbers 1–12). Dots represent sites from stable Istria, squares from imbricated Adria. Large grey dots = moderately tilted strata; Large black dots = subhorizontal strata; Small black dots = localities with no palaeomagnetic resuls due to weak NRM; Small black squares = localities with no palaeomagnetic results due to weak NRM. Large grey squares = good palaeomagnetic results.

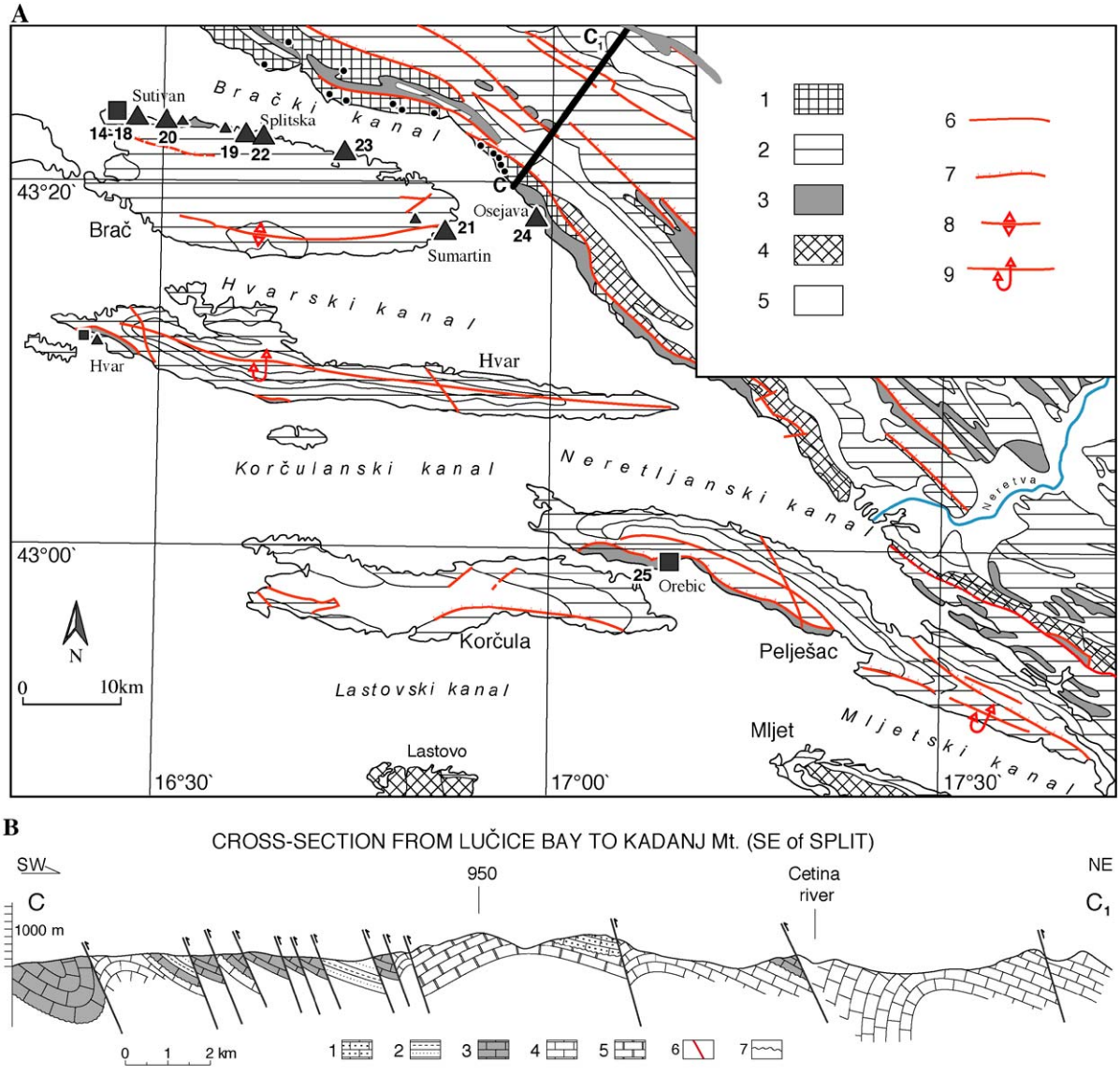


Fig. 3. Palaeomagnetic sampling area 4 (Fig. 1) with the palaeomagnetic sampling localities. (A) Geological sketch map of imbricated Adria in Central Dalmatia (after the Geological map of former Yugoslavia 1:500 000) with sampling sites (squares, triangles and numbers 14–25) and the location of the geological cross-section (C–C1). Key: Large squares = Eocene strata with good palaeomagnetic results; Large triangles = uppermost Maastrichtian strata with good palaeomagnetic results; Small squares = weakly magnetized Eocene strata; Small triangles = weakly magnetized uppermost Maastrichtian strata; Dots = sampling sites of Kissel et al. (1995). Key to geology: 1 = Upper Eocene (Priabonian) flysch; 2 = Middle Eocene (Lutetian–Bartonian) foraminiferal limestones; 3 = Upper Cretaceous limestones; 4 = Lower Cretaceous limestones; 5 = Triassic–Jurassic limestones; 6 = faults; 7 = reversal faults; 8 = axis of anticline or syncline; 9 = axis of overturned syncline or anticline. (B) Geological cross-section from Lucice bay to Kadanj Mt. (SE of Split): data taken from Marinčić et al. (1976). 1 = Oligocene breccia; 2 = Upper Eocene (Priabonian) flysch; 3 = Middle Eocene (Lutetian) foraminiferal limestone; 4 = Upper Cretaceous limestone (Cenomanian–Maastrichtian); 5 = Lower Cretaceous limestone; 6=reversal faults; 7=transgressive boundary.

1924; Petković, 1958; Aubouine, 1974; Grubić, 1975; Dimitrijević, 1982; Pamić et al., 1998; Herak, 1986, 1999; Tari, 2002). Authors generally agree that the eastern Adriatic coast consists of a relatively autochthonous platform, which is overthrusted by nappes, but disagree about the paleogeography and tectonic evolution. For instance, Pamić et al. (1998) assumes a uniform Adriatic–Dinaridic carbonate platform, while Herak (1986, 1999) distinguishes an Adriatic platform and a Dinaric platform. Biostratigraphic data (from the K/T boundary to the Middle–Late Eocene) confirm the subdivision suggested by Herak (1986, 1999). Tari (2002, Fig. 1) distinguishes a gently tectonized Adriatic carbonate platform (Istria), with an imbricated margin (imbricated Adria) and the Dinaric nappe pile.

In the study area, a stratigraphic gap is often observed between the Cretaceous and the Eocene. Locally, sedimentation is continuous across the K/T boundary (Brač island; Gušić and Jelaska, 1990). In areas 1–3 of Fig. 1, Valangian to Santonian, (Matičec et al., 1996), and in Central Dalmatia (area 4, Fig. 1) Maastrichtian (Gušić and Jelaska, 1990; Ćosović; et al., 1994) shallow water platform carbonates are trangressively overlain by Eocene brackish to lagoonal limestones (Liburnia Formation) that pass into Foraminiferal limestones. Marly limestones are transitional beds between older platform carbonates and younger flysch, which indicates the deepening of the sedimentary basin due to tectonic disintegration of the platform, accompanied by sea level rise (Cuisian/Lutetian to Priabonian).

Several tectonic events are recognized during the time period considered in this paper. The first is manifested in an unconformity between Late Cretaceous and Paleogene strata (Pamić et al., 1998). The strongest tectonic movements are responsible for the southwesterly vergence of the structures of the eastern Adriatic coast. According to Pamić et al. (1998) this deformation took place in Latest Eocene–Early Oligocene times. Tari (2002) suggests that folding and thrusting started in the Oligocene, when collision and progressive underthrusting of Adria below the Dinaric carbonate platform created imbricated structures at the margin of Adria; the process continued during the Miocene and after. The typical "Dinaric structures" striking NW–SE were refolded during neotectonic movements (Marinčić, 1997) producing W–E orient-

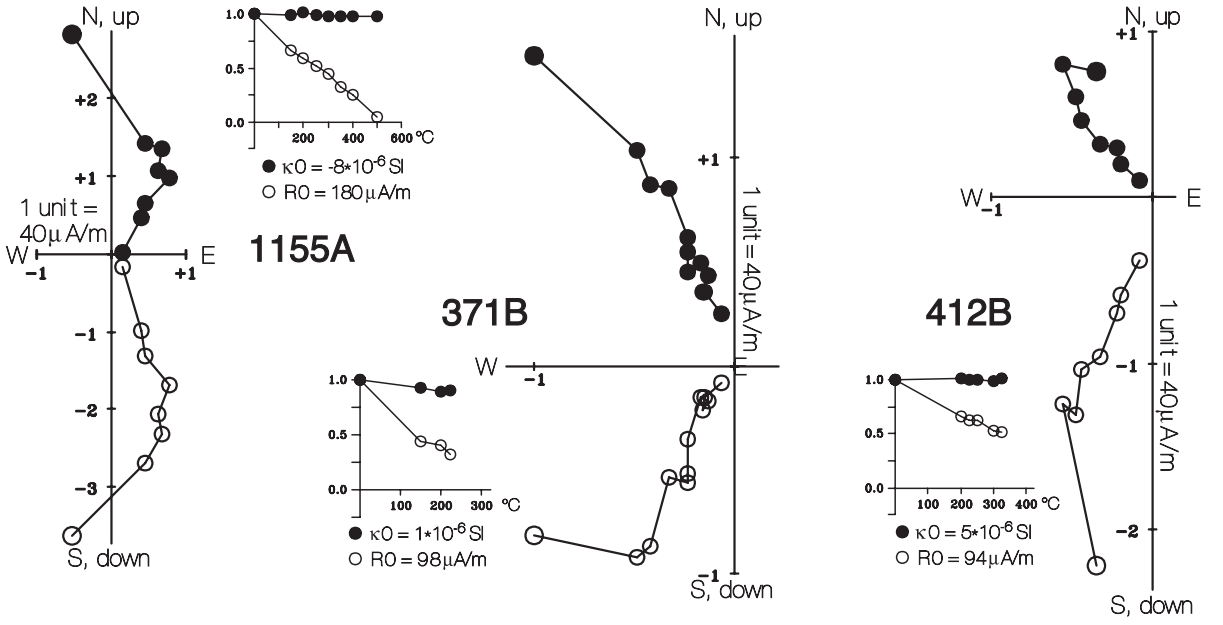


Fig. 4. Typical demagnetization curves for areas 1–3. Original Zijderveld diagrams and intensity/susceptibility versus temperature curves. In the Zijderveld diagrams, full/open circles: projection of the NRM in the horizontal/vertical plane; in the others susceptibility: dots, NRM intensity: circles. Specimens 371B and 412B were AF demagnetized following thermal demagnetization, in max AF field of 25 and 50 mT, respectively.

ed structures which are most conspicuous in the Central Dalmatian islands.

3. Palaeomagnetic sampling

We collected fossiliferous rocks from 58 sites avoiding strata which were rich in fossils, karstified or weathered. At some localities, we drilled sections spanning through millions of years and representing different lithologies (Fig. 2, locality 2 and Fig. 3A, sites 14–18). The cores were oriented in the field using a magnetic compass. Samples for biostratigraphic checking (based on larger foraminifers in the Maastrichtian and Paleogene carbonates, and on planktonic foraminifers in the marls) were also taken.

From stable Istria and from the imbricated area situated to the N of it (Fig. 1, areas 1 and 2), Lower Eocene through Bartonian shallow water limestones, marly limestones (sometimes with glauconite) and marl beds from flysch were collected. Limestones of Cuisian age were sampled at three localities from area 3 (Fig. 1). In Central Dalmatia (imbricated Adria, Fig. 1, area 4), most of our samples are from latest Maastrichtian strata (about 65–66 Ma, when referred to the polarity time scale of Gradstein et al., 1994), with highly variable dips (8–51°).

4. Laboratory measurements and results

The natural remanent magnetization (NRM) and the susceptibility of each specimen was measured in the Palaeomagnetic Laboratory of Eötvös Loránd Geophysical Institute of Hungary using JR-5A and JR-4 spinner magnetometers and a KLY-2 Kappabridge.

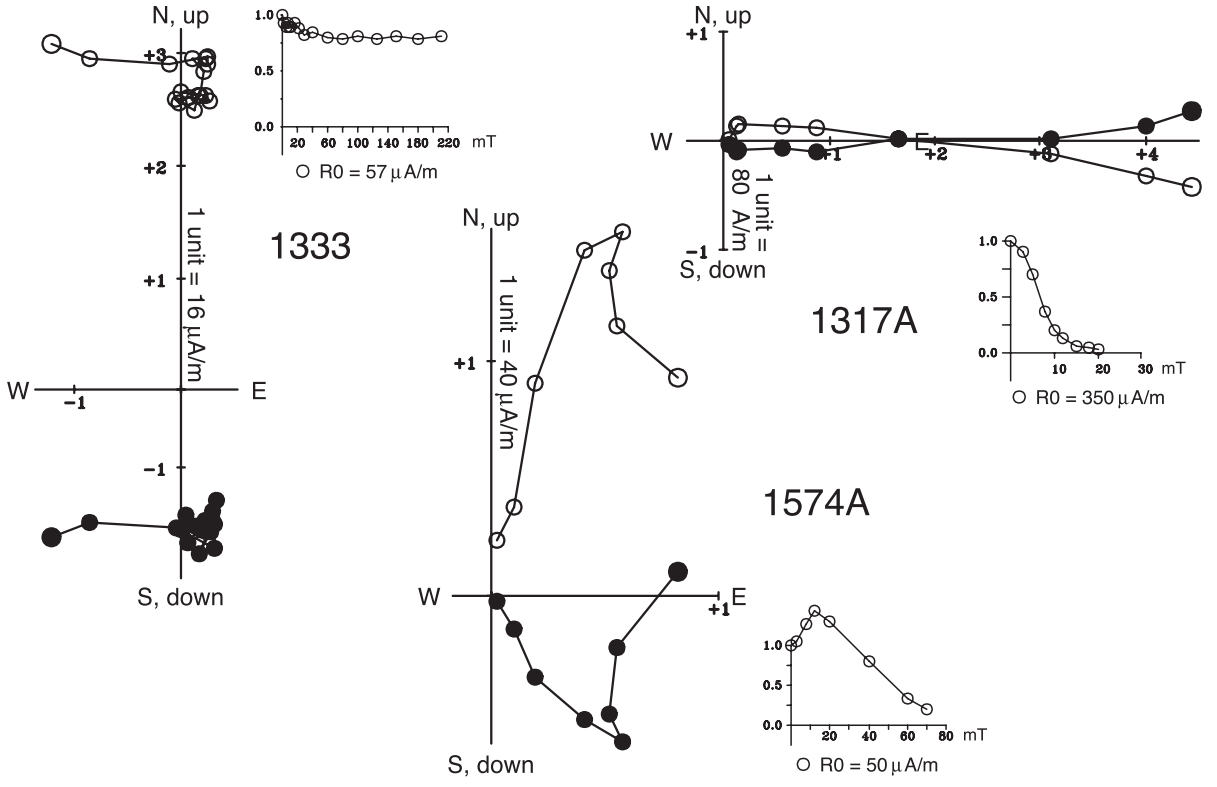


Fig. 5. Typical demagnetization curves for area 4. Original Zijderveld diagram and intensity versus demagnetizing field curves. In the Zijderveld diagrams, full/open circles: projection of the NRM in the horizontal/vertical plane; in the others susceptibility: dots, NRM intensity: circles. Note that AF demagnetization up to 220 mT was reducing the NRM intensity of specimen 1333 by about 10% (main carrier of the NRM is hematite), thus thermal demagnetization was needed to demagnetize the NRM.

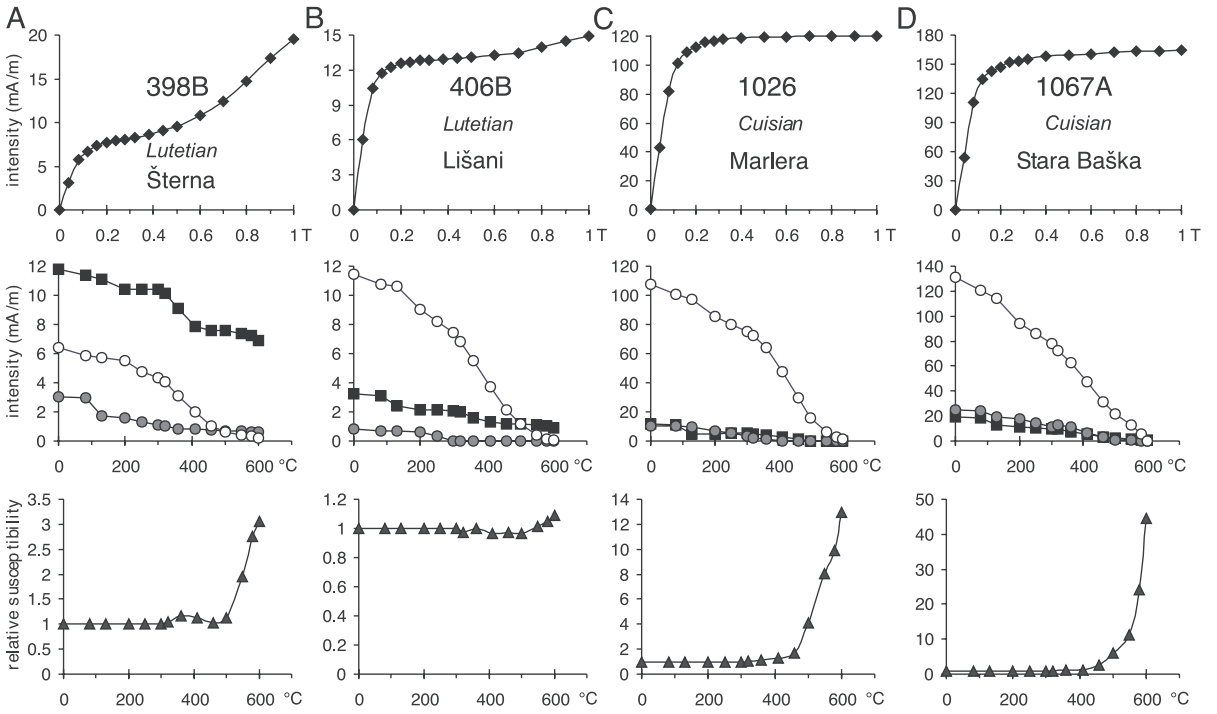


Fig. 6. Magnetic mineralogy of the sediments from areas 1–3 (Fig. 1). Typical IRM acquisition curves (upper row), the behaviour of the three component IRM (Lowrie, 1990) on thermal demagnetization (second row) and the change in susceptibility on heating (lowermost row). The hard (squares), the medium hard (dots) and soft (circles) components of the composite IRM were acquired in fields of 1.0, 0.36 and 0.2 T, respectively. Note that the dominant magnetic mineral is magnetite in the subhorizontal localities of stable Istria (B and C) while it is hematite in the moderately tilted localities (A).

Specimens with NRM intensities at least five times higher than the noise level of the spinner magnetometers (3×10^{-6} A/m) were demagnetized in small increments, either by alternating field (AF) or thermal demagnetization (Fig. 4). In the latter case, susceptibility was monitored in order to detect possible mineralogical changes on heating. The results were subjected to principal component analysis (Kirschvink, 1980). Magnetic mineralogy was studied by isothermal remanent magnetization acquisition (IRM) experiments and by the stepwise thermal demagnetization of three-component IRMs (Lowrie, 1990).

Although the NRM was weak to very weak, we obtained well-defined components of magnetizations at 25 sites. AF demagnetization was efficient in most cases, but samples from certain sites could not be demagnetized up to 0.21 Tesla (Fig. 5) and thermal demagnetization was necessary. In such exceptional cases, hematite was an important magnetic mineral,

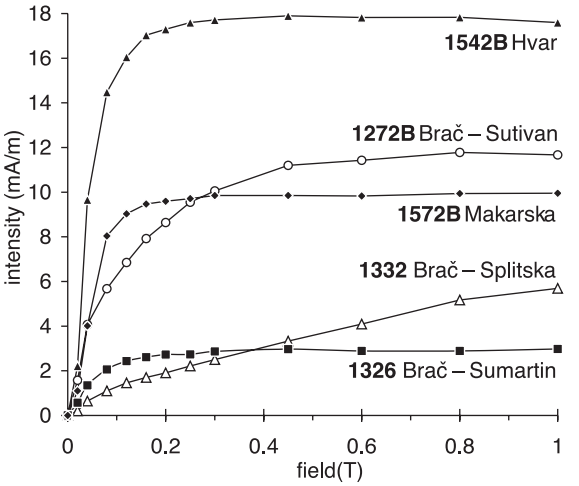


Fig. 7. Typical IRM acquisition curves for area 4 (Fig. 1). Note that specimen 1332 with hematite represents the same site as 1333 in Fig. 5.

Table 1
Stable Istria (area 1 in Fig. 1) Palaeomagnetic directions with statistical parameters

Locality	Lithology age	*n*/no	*D*°	*I*°	*k*	α_{95}°	D_C°	I_C°	*k*	α_{95}°	Dip
(1) Marlera 1223–1239	limestone *Cuisian*	11/17	141	−62	32	8	142	−60	32	8	334/2
(2) Pican 367–396	marl *Lutetian*	10/23	331	+54	14	13	331	+54	14	13	horizontal
(3) Mali Mlun 1857–1872	flysch *Lutetian–Bartonian*	11/16	166	−54	97	5	166	−52	97	5	348/2
(4) Livade 1783–1790	limestone *Cuisian*	7/8	346	+51	26	12	346	+51	26	12	horizontal
(5) Ragancini–Lisani 406–418	limestone *Lutetian*	8/13	151	−62	40	9	158	−66	40	9	110/5
Overall paelomagnetic mean direction		5/5	336	+57	114	7	337	+57	95	8	
(6) Murine 1847–1856	limestone *Cuisian*	5/10	337	+41	108	8	348	+37	108	8	55/14
(7) Sterna 397–405	limestone *Lutetian*	7/9	161	−49	83	7	151	−26	83	7	308/26
(8) Sv. Ivan 1156–1163	limestone *Cuisian*	7/8	161	−40	20	14	174	−41	20	14	80/15
Overall paelomagnetic mean direction		8/8	337	+52	72	7	340	+49	30	10	

Key: *n*/no: number used/collected samples (the samples are independently oriented cores) or sites/localities used/collected; *D*, *I* (D_C, I_C): declination, inclination before (after) tilt correction; *k* and α_{95}: statistical parameters (Fisher, 1953).

Table 2
The NW part of imbricated Adria (areas 2–3 in Fig. 1) Palaeomagnetic directions with statistical parameters

Locality	Lithology age	*n*/no	*D*°	*I*°	*k*	α_{95}°	D_C°	I_C°	*k*	α_{95}°	Dip
Istria											
(9) Prešnica 25–31	marl *Lutetian*	7/7	98	−22	35	10	131	−51	35	10	57/49
(10) Kubed 1–8	marl *Lutetian*	8/8	318	+63	20	13	1	+50	20	13	45/30
(11) 9–14	marl *Lutetian*	5/6	312	+56	71	9	350	+47	71	9	45/30
(12) Lukini 1771–1781	marl *Bartonian*	11/12	92	−28	10	15	104	−44	10	15	52/23
Overall mean direction		4/4	288	+46	9	33	326	+52	12	28	
Krk											
(13) Stara Baska 1153–1159 1264–1269	limestone *Cuisian*	9/13[a]	329	+44	22	11	321	+12	22	11	301/35
Pretilting and overprint combined		7/13[b]	36	+35	77	7	354	+44	77	7	
		16/13					340	+45	23	8	
Istria + Krk overall mean direction		5/5					330	+50	15	20	

Key as for Table 1.
[a] Overprint remanence.
[b] Primary remanence.

Table 3
The SE part of imbricated Adria (Central Dalmatia, area 4) Palaeomagnetic directions with statistical parameters

Locality	Lithology age	*n*/no	*D*°	*I*°	*k*	α_{95}°	D_C°	I_C°	*k*	α_{95}°	Dip
Brač											
Sutivan–Borak section											
(14) 1270–1273	limestone *Maastrichtian*	4/4	159	−53	168	7	173	−59	168	7	30/19
(15) 1748–1752	limestone *Maastrichtian*	5/5	330	+58	30	13	322	+48	30	13	6/18
(16) 1282–1287	limestone *Maastrichtian*	3/6	322	+79	100	12	353	+66	100	12	16/16
(17) 1288–1293	limestone *Maastrichtian*	4/6	157	−43	101	7	157	−35	101	7	340/8
(18) 1317–1320	marly limestone *Cuisian*	4/4	117	−44	322	5	129	−28	322	5	348/22
(19) Postire 1726–1731	limestone *Maastrichtian*	5/6	347	+56	75	9	350	+49	75	9	8/8
(20) Mirca 1755–1762	limestone *Maastrichtian*	6/8	335	+69	68	8	352	+61	68	8	27/11
(21) Sumartin 1321–1326	limestone *Maastrichtian*	4/6	339	+55	30	17	331	+66	30	17	176/12
(22) Splitska 1327–1334	limestone *Maastrichtian*	8/8	154	−52	87	6	162	−34	87	6	2/20
(23) Pučišče 1930–1942	limestone *Maastrichtian*	4/13	331	+51	62	12	342	+36	59	12	variable
Makarska											
(24) Osejava section 1571–1577	limestone *Maastrichtian*	7/7	129	−57	92	6	165	−38	92	6	28/36
Peljesac											
(25) Orebic section 1550–1570	limestone *Cuisian*	4/21	322	+72	130	8	354	+23	130	8	10/51
Overall mean direction, area 4		12/12	329	+58	36	7	340	+46	20	10	

Key as for Table 1.

while the usual magnetic mineral was magnetite (Figs. 6 and 7). As a result of principal component analysis and statistical evaluation, we obtained the site mean directions shown in Tables 1–3.

5. Discussion and conclusions

Post-Cretaceous outcrops from stable Istria are well-documented, since this area has been the subject of intensive biostratigraphical studies (Drobne, 1977, 1979; Drobne and Pavlovec, 1991; Ćosović and Drobne, 1998). For the palaeomagnetic study, we drilled all localities that seemed lithologically suitable to yield positive results. As Fig. 2 shows, the total number is 11, of which three were too weakly magnetized to provide satisfactory demagnetization curves (small black dots, Fig. 1). Of the remaining eight localities, five represent subhorizontal beds (large black dots) and three moderately tilted strata (large grey dots). Subhorizontal localities yield very consistent palaeomagnetic mean directions with declinations significantly rotated in the counterclockwise (CCW) sense (Fig. 8A). In these samples, magnetite is the dominant magnetic mineral (Fig. 6C). The palaeomagnetic directions of the remaining three localities are clustered better before than after tilt corrections, close to the overall-mean palaeomagnetic direction for the subhorizontal strata. The dominant magnetic mineral in this group is hematite (Fig. 6A). Although a fold test cannot be applied to the group of subhorizontal localities, they are likely to carry primary remanence, since the secondary remanence seems to be connected to the presence of abundant hematite. An additional evidence for primary magnetization in the subhorizontal strata is that at locality 1, where we

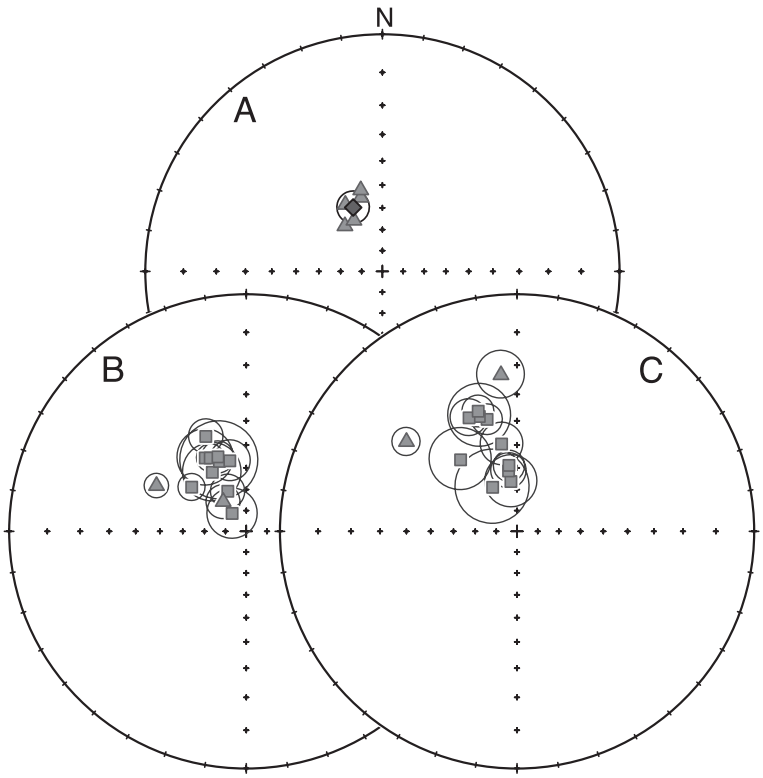


Fig. 8. Summary of palaeomagnetic locality (site) mean directions from (A) stable Istria (area 1, Fig. 1 subhorizontal beds) and from Central Dalmatia (area 4, Fig. 1) for uppermost Maastrichtian (squares)–Eocene (triangles) platform carbonates before (B) and after (C) tilt corrections. Stereographic projections.

could take samples from subhorizontal as well as from tilted beds, the former yielded good palaeomagnetic signal while the latter had no signal exceeding the noise level of the magnetometers. In other words, the NRM that was destroyed by gentle tectonization was not replaced by a newly acquired one.

Reference poles for Africa, published by different authors (e.g. Besse and Courtillot, 1991; Muttoni et al., 2001), significantly differ from the paleopole calculated for the subhorizontal strata from stable Istria (Fig. 9A), clearly suggesting that the latter rotated counterclockwise with respect to the former by an angle of about 27°.

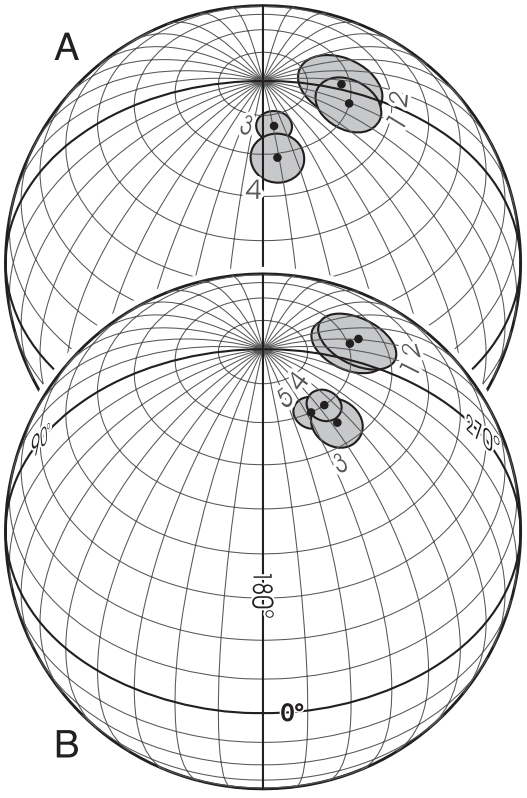


Fig. 9. Paleopoles (A) for the subhorizontal strata from stable Istria (1) and for all localities for stable Istria (2) compared with reference poles of similar age for Africa (3: 50 Ma) (Besse and Courtillot, 1991), (4: 45 Ma) (Muttoni et al., 2001). (B) For latest Maastrichtian strata from Central Dalmatia, postfolding remanence (1), for latest Maastrichtian plus Cuisian strata postfolding remanence (2) compared with poles of similar age for Africa (3: 60–80 Ma) (Besse and Courtillot, 1991) based on direct African poles (4: 70 Ma) syntethic pole, (5: 60 Ma) (Muttoni et al., 2001). All paleopoles are plotted with α_{95}.

In areas 2 and 3 (Fig. 1, imbricated Adria), 21 sites were tested from nine localities, but few of them (Table 2) yielded tectonically useful palaeomagnetic direction (rejected sites are mostly too weakly magnetic or had directions aligned with the present local geomagnetic field before tilt correction). The overall mean palaeomagnetic direction based on the data of Table 2 exhibits smaller between-site scatter after than before tilt correction. The improvement is insignificant, the overall-mean palaeomagnetic directions are practically the same in both cases, and similar to that of stable Istria.

In area 4 (Fig. 1), several sites were too weakly magnetic to yield results, but none of them was seriously overprinted in the present field. The magnetic mineral was either soft magnetite alone or accompanied by hematite (Fig. 7). The overall mean palaeomagnetic direction calculated from the uppermost Maastrichtian results has the maximum Fisher precision parameter k at around 30% of untilting, without being significant at the 95% probability level. The McFadden (1990) fold test, which is very effective in this case (the angle between the in situ overall mean and the unfolded overall mean is small relative to the movements of the individual site-mean directions), gives a definite answer: ξ_1 is lowest at zero unfolding (0.360) and the test is highly significant (critical value is 3.685, at complete unfolding $\xi_1 = 4.576$). The situation is similar when the two Eocene sites (directions are similar to those of the Maastrichtian ones) are added (ξ_1 min 1.720, critical value is 4.036, at full unfolding $\xi_1 = 5.876$) These results (Fig. 9) suggest that Central Dalmatian and stable Istrian carbonate platforms rotated CCW with respect to Africa during the Tertiary, in a highly coordinated manner.

In Central Dalmatia, the flysch studied by Kissel et al. (1995) lies directly on the platform studied by us, and both belong to imbricated Adria. However, the magnetization of the platform (showing CCW rotation) is younger than that of the flysch (showing no rotation?), having been acquired in the Oligocene, due to migration of fluids or even later, in the Miocene due to migration of hydrocarbon (Barić and Tari, 2001). Field inspection and also the section in Fig. 3B shows that the flysch experienced contraction between two, more rigid units consisting of platform carbonates

(Dinaric nappes over-ride the flysch). Contrary to the moderate tectonization of area 4 (Fig. 1), the flysch is heavily tectonized at some of the localities used by Kissel et al. (1995) for tectonic interpretation. At such localities, the strike is strongly deviated from the general Dinaric trend of 120–300° which was interpreted by Kissel et al. (1995) as the consequence of oroclinal bending. We think that the oroclinal correction of Kissel et al. (1995) is problematic for the following reasons: (1) The deviation from the general trend is not systematic, but varies locally, thus the deviations must have been caused by tilting about oblique axes or refolding (folding in the flysch is partly synsedimentary; Marinčić, 1981; Marjanac, 1990, 1996); (2) Magnetic lineation (Speranza, 1995) was used as a proxy for strike, although there are large deviations between strike measured in the field and the magnetic lineation for the same locality (sedimentary transport as a cause of the lineation was not considered); (3) Oroclinal corrections were calcu-

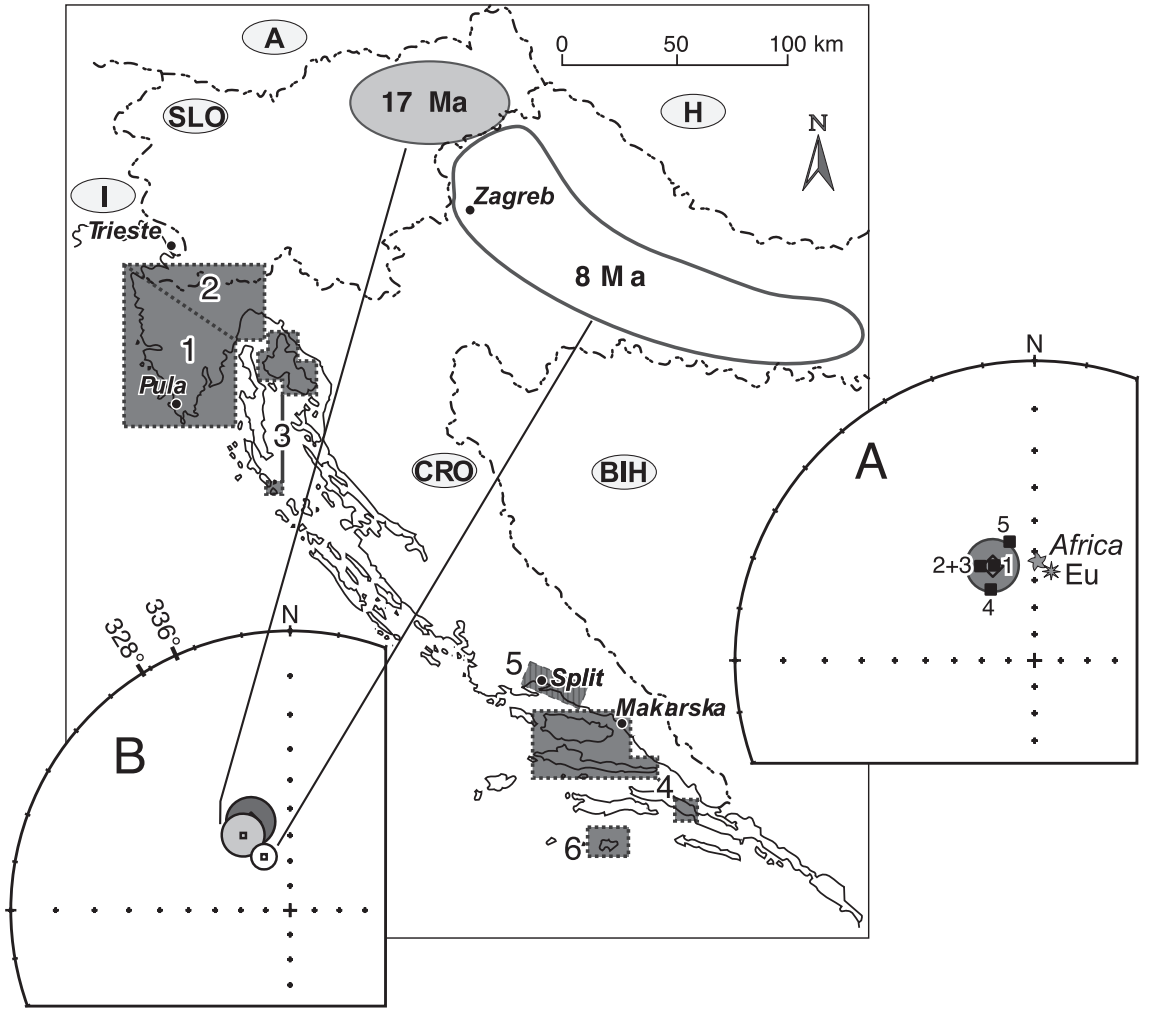


Fig. 10. Comparison between area mean Teriary palaeomagnetic directions from the eastern Adriatic coast, Northern Slovenia and Northern Croatia, respectively. In (A), reference directions in an African and European framework were calculated from poles compiled by Besse and Courtillot (1991) for 45 Ma, the regional mean with dark grey α_{95} was calculated from area means 1–5. (B) compares area mean of 1–5 of (A) and those characterizing the youngest rotation of NW Croatia (white, dominant age of the studied rocks is about 8 Ma) and NE Slovenia (light grey, dominant age of the studied rocks is about 17 Ma).

lated as the difference between individual site mean lineation directions and the general Dinaric trend defined by structural observation (120–300°), and not the general lineation direction, which is 110–290°. If we disregard the highly questionable oroclinal corrections and reject two localities with palaeomagnetic directions very close to the direction of the present local geomagnetic field in the geographic system, the result will be Dec = 348°, Inc = 46°, $\alpha_{95} = 6°$. This overall mean palaeomagnetic direction for the Central Dalmatian flysch is consistent with the results of the present study (Fig. 10A).

Lack of rocks younger than the Eocene in our study area makes it impossible to obtain direct constraints for the exact age of the rotation(s). However, the rotation of Adria may be related to strong deformation and block rotation of late Miocene (Pliocene?) age north of the microplate. Examples are the Mura–Zala depression (Fodor et al., 1998; Márton et al., 2002a), NW Croatia (Tomljenović and Csontos, 2001; Márton et al., 1999, Márton et al., 2002b) the Transdanubian Range in Hungary (Márton and Fodor, 2003) or more generally, in the Pannonian Basin (Bada et al., 1999).

The results of the present study are also relevant to the problem of the shape of the Dinarides and the Hellenides. While agreeing that the Scutari–Peć line was a significant boundary between them, Márton (1987) suggested that the Scutari–Peć line was the contact zone between two microplates, rotating in opposite senses, while Kissel et al. (1995) put forward an alternative model, in which the structures of the eastern Adriatic coast were bent from the CCW rotated NW part through the non-rotated SE segment, and the CW rotated Hellenides (e.g. Horner and Freeman, 1983; Kissel et al., 1985, 1995; Mauritsch et al., 1995). The present study supports the model of two blocks, in which a Dinaric (sensu lato) and a Hellenic microplate rotate in opposite senses.

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